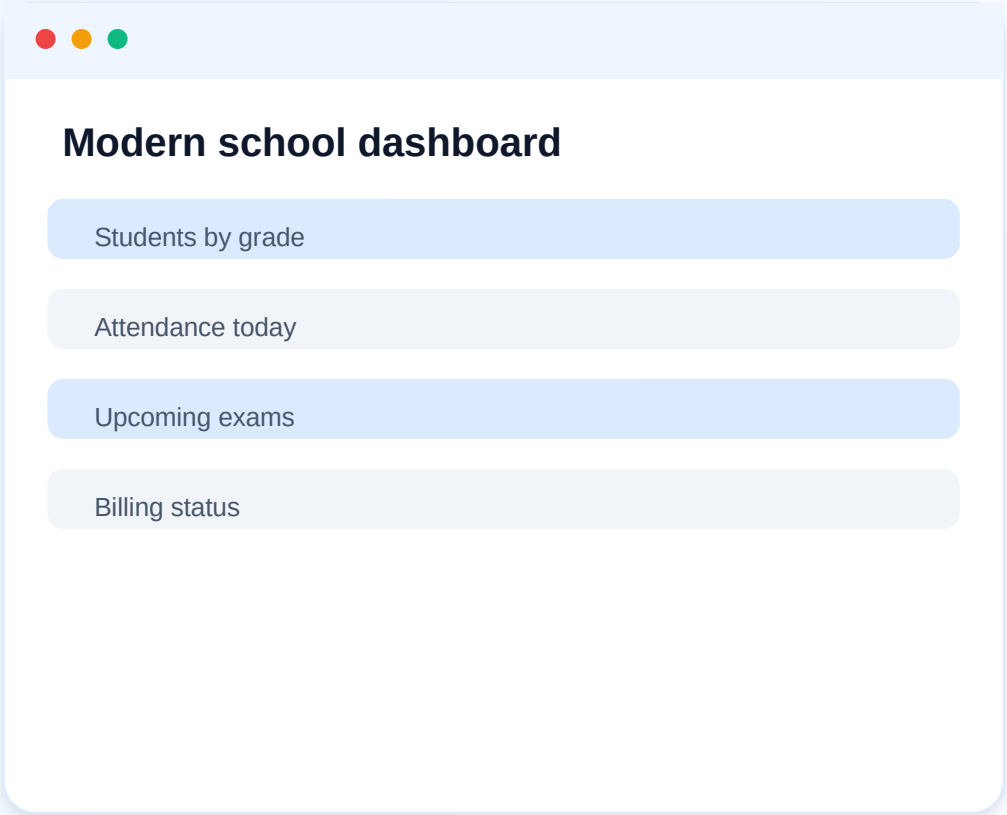
SOFTWARE ENGINEERING PROJECT

Campus ERP - School Management System

A Multi-Tenant SaaS Platform for Complete School Management

Logo placeholder

Team / project information



Demo: campuserpt.vercel.app

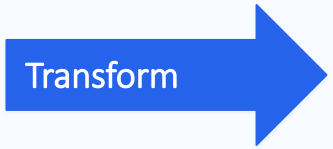
PROBLEM

Schools need one source of truth

Traditional processes create duplicated work, delayed decisions, and poor communication.

Traditional workflow

- Paper records
- Separated departments
- Slow reporting
- Hard to scale



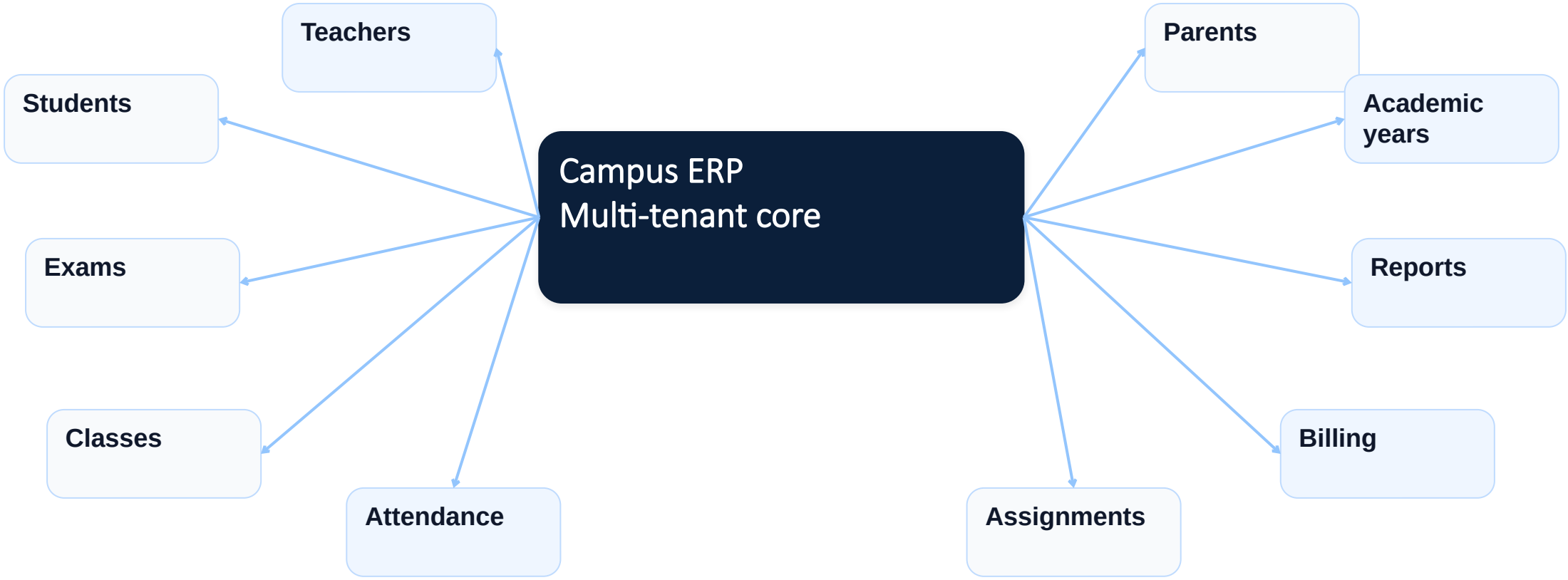
Digital ERP platform

- Shared data model
- Role-based access
- Live academic operations
- Multi-school scale

OVERVIEW

Campus ERP unifies the school operating model

A complete SaaS layer for academic, people, communication, and billing workflows.



GOALS

The system is built for secure growth

The design targets SaaS scale while preserving school-level privacy and operational completeness.

Sc

Scale

Support many schools on one platform

Is

Isolation

Keep tenant data separated by design

Ac

Access

Use roles and permissions for every workflow

Li

Lifecycle

Manage the academic year from setup to reports

Fu

Future

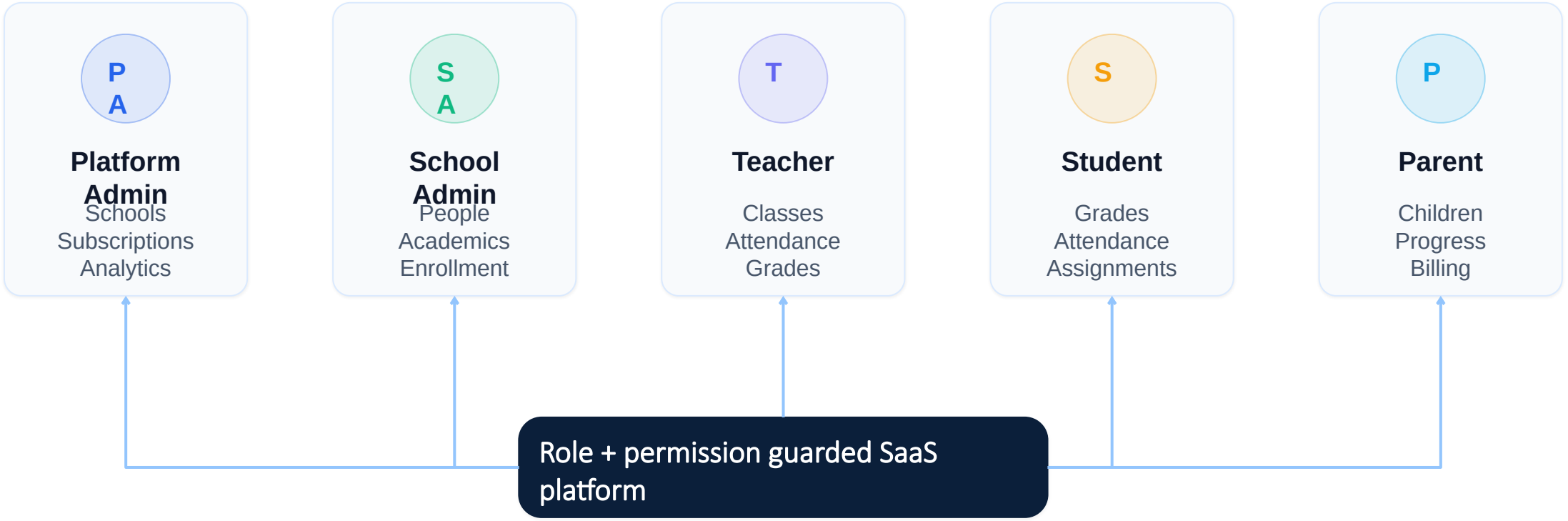
Leave room for mobile apps and analytics

Engineering objective: one product foundation that can grow across tenants, modules, and devices.

USERS

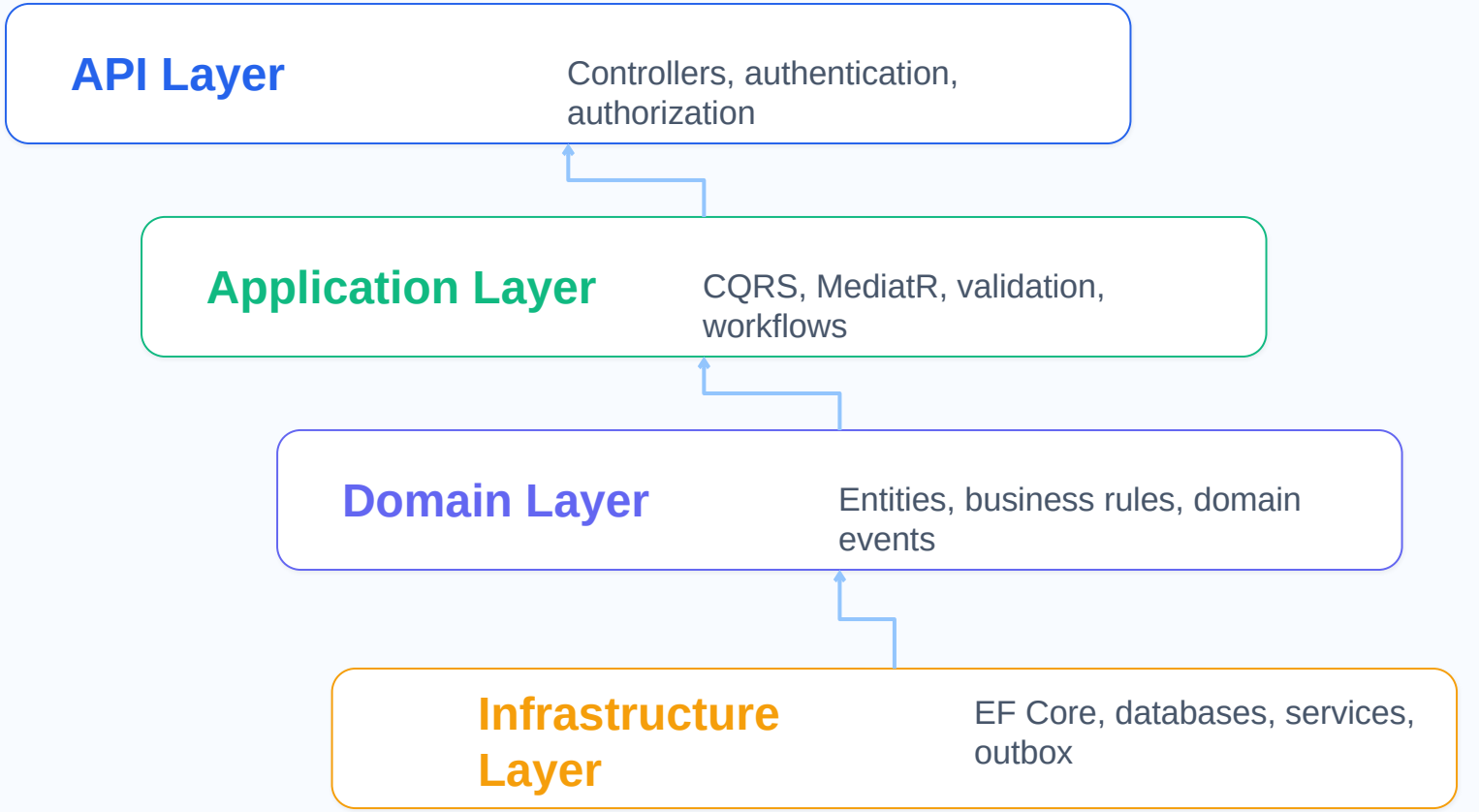
Five roles cover platform, school, classroom, and family needs

Each actor sees only the workflows and data they are allowed to access.



Clean Architecture keeps business logic independent

Controllers stay thin while application handlers coordinate domain rules and infrastructure.

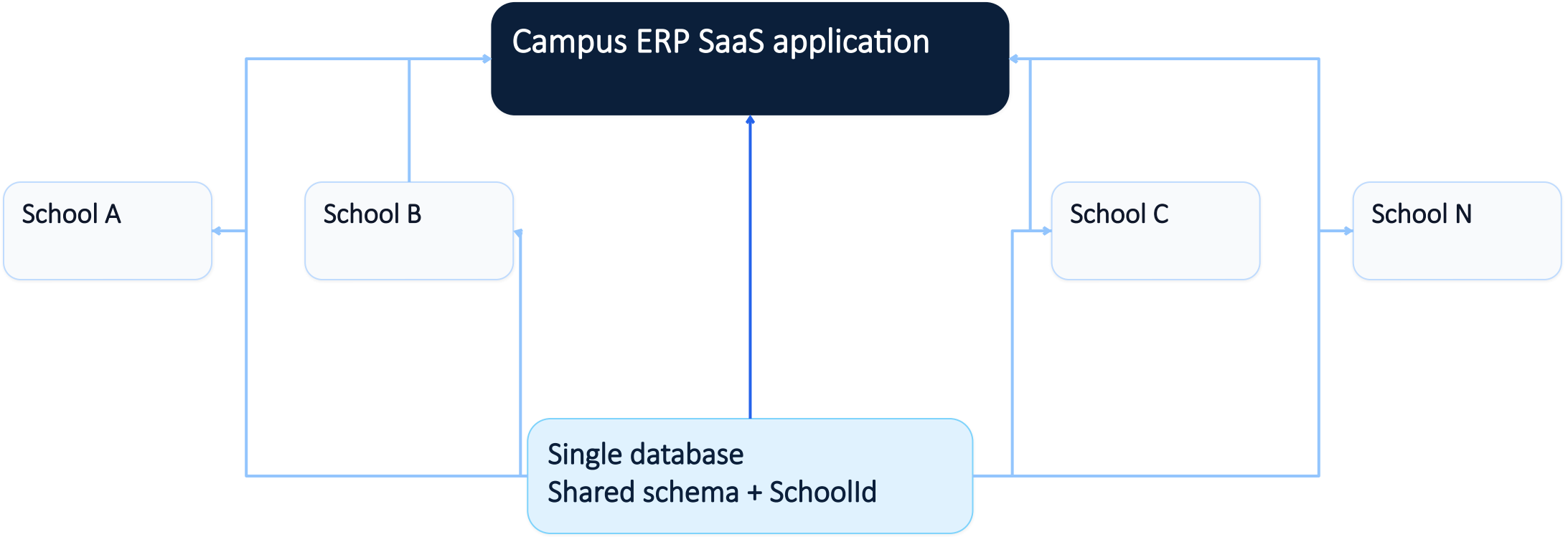


Tech stack

ASP.NET Core
EF Core + SQL Server
MediatR
FluentValidation
JWT

One SaaS platform securely serves many schools

A shared schema uses SchoolId and query filters to isolate tenant data.



Isolation controls: JWT tenant context + tenant authorization behavior + EF Core global query filters

DATA MODEL

The ERD separates platform ownership from school operations

Platform entities manage tenants and billing; application entities manage academic workflows.

Platform	People	Academic	Operations
School	**Student**	**Academic Year**	**Enrollment**
Campus	Teacher	Term	Attendance
Subscription	Parent	Classroom	Assignments
Invoice	School Admin	Subject	Exams
Payment			Results

ERD reference: app.eraser.io/workspace/wsKQPubwZL5qvcdTG50x

FEATURES

Core modules cover the daily school lifecycle

The platform connects secure access, management data, academic setup, and daily operations.

AU

Authentication

JWT login, refresh tokens, role and permission claims.

SC

School Management

Student, teacher, parent, guardian, and school records.

AC

Academic Management

Academic years, terms, grades, rooms, classrooms, schedules.

DA

Daily Operations

Attendance, assignments, exams, report cards, notifications.

SECURITY

Security is layered across identity, authorization, and data access

The design protects both user identity and tenant boundaries.

1	**JWT authentication**	Signed access tokens and refresh tokens
2	**Role-based authorization**	Platform admin, school admin, teacher, student, parent
3	**Permission policies**	Fine-grained claims guard API endpoints
4	**Tenant isolation**	SchoolId context and EF global query filters
5	**Audit + soft delete**	Track changes and preserve operational history
6	**Concurrency protection**	Protect writes in multi-user workflows

TECHNOLOGY

The stack balances modern frontend UX with enterprise backend patterns

Each layer uses mainstream tools that fit the SaaS architecture.

1 Backend

- ASP.NET Core
- Entity Framework Core
- SQL Server
- MediatR
- FluentValidation

2 Frontend

- React
- TypeScript
- Tailwind CSS
- Ant Design

3 Infrastructure

- Docker
- Cloud deployment
- Email services
- Outbox processing

ROADMAP

The roadmap expands from foundation to scale

Each phase adds a product layer on top of the secure SaaS core.



Phase 1

Core SaaS foundation

Phase 2

Attendance and daily operations

Phase 3

Exams, reports, notifications

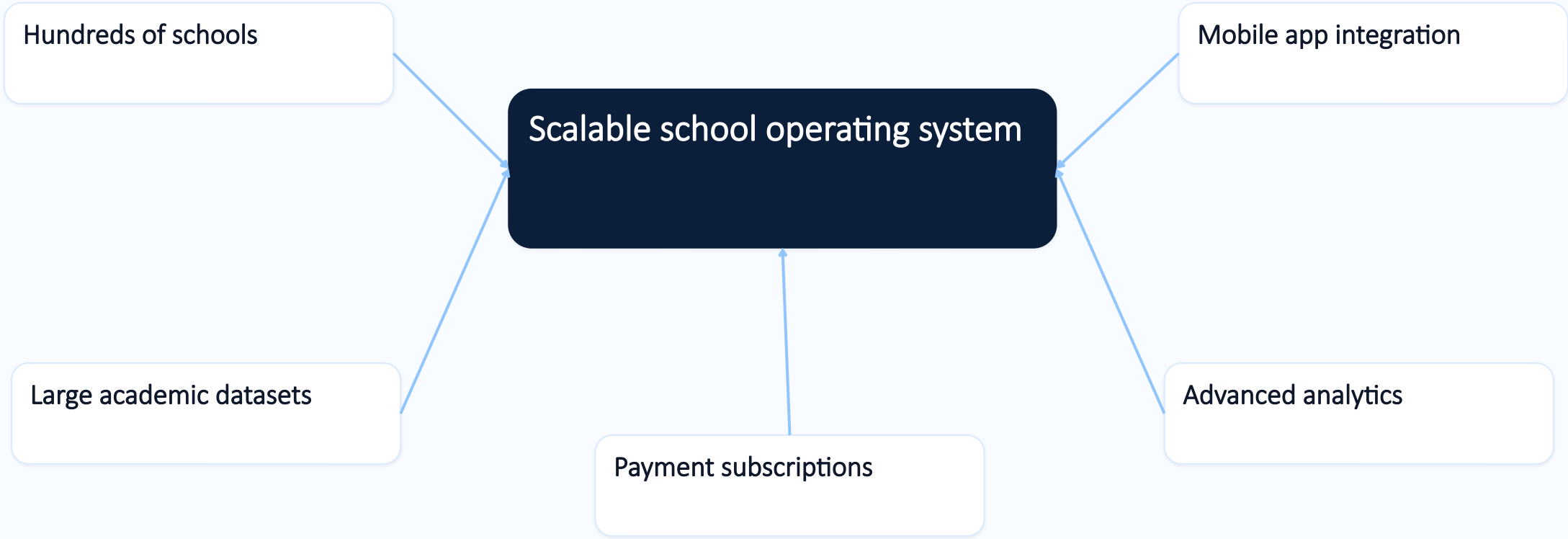
Phase 4

Billing system and scaling

VISION

Campus ERP can grow into an intelligent school platform

The same architecture can support larger datasets, mobile apps, analytics, and subscription growth.



Campus ERP transforms school management into a secure, scalable, and intelligent digital platform.

Thank you